

Mathematics

Stage 8

Paper 2

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

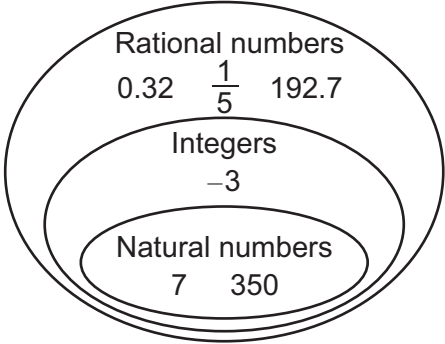
Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer The graph line passing through a point implies the point even though there is no cross	A horizontal line and vertical line from the axes meeting at the required point

Question	Answer	Marks	Part Marks	Guidance
1	Any two integers satisfying $x < 8$	1		e.g. ($x =$) -7 or 0 or 3
2	6	1		Accept 0
3	$a \quad t \quad \textcircled{u} \quad v$	1		Accept any clear indication.
4	0.35	1		Accept equivalent fractions or percentage.
5	All squares are rectangles. ☒ All rhombuses are squares. ☐ All parallelograms are rhombuses. ☐ All rectangles are parallelograms. ☒	1		Accept any clear indication. Both answers correct for the mark.
6	Any correct reason to do with ease / accuracy, e.g. - The whole numbers have been kept separate (so the calculations are easier to do). - The fractions are easier/smaller to work with (so less chance of an arithmetic error).	1		
7	$C = 35 + 20d$	1		Accept equivalents, e.g. $20 \times d + 35 = C$ Accept with units included, e.g. $\$C = \$35 + \$d20$ Do not accept an expression.

Question	Answer	Marks	Part Marks	Guidance									
8(a)	Input (x) $\rightarrow$ $\boxed{\times 2}$ $\rightarrow$ $\boxed{-4}$ $\rightarrow$ Output (y)	1											
8(b)	<table><tr><th>Input</th><th>Output</th></tr><tr><td>(−3.5)</td><td>−11</td></tr><tr><td>7.5</td><td>(11)</td></tr></table>	Input	Output	(−3.5)	−11	7.5	(11)	2	Award 1 mark for each correct answer.				
Input	Output												
(−3.5)	−11												
7.5	(11)												
9	792 000 0.40	2	Award 1 mark for each correct answer.	Do not accept 792 Do not accept 0.4									
10(a)	55	2	Award 1 mark for $\frac{5}{14} \times 154$ or equivalent or for finding one share is worth 11										
10(b)	<table><tr><td></td><th>Ratio of red to black pens</th><th>Proportion of red pens</th></tr><tr><td>Box 1</td><td>(2 : 7)</td><td>$\frac{2}{9}$</td></tr><tr><td>Box 2</td><td>3 : 7</td><td>$\left(\frac{3}{10}\right)$</td></tr></table>		Ratio of red to black pens	Proportion of red pens	Box 1	(2 : 7)	$\frac{2}{9}$	Box 2	3 : 7	$\left(\frac{3}{10}\right)$	2	Award 1 mark for each correct answer.	Accept any equivalent fractions and ratios.
	Ratio of red to black pens	Proportion of red pens											
Box 1	(2 : 7)	$\frac{2}{9}$											
Box 2	3 : 7	$\left(\frac{3}{10}\right)$											

Question	Answer	Marks	Part Marks	Guidance
11(a)(i)	$\frac{1}{5}$	1		Accept any equivalents, e.g. $\frac{3}{15}$ or 0.2 or 20%
11(a)(ii)	$\frac{3}{5}$	1		Accept any equivalents, e.g. $\frac{9}{15}$ or 0.6 or 60%
11(b)	D	1		
12(a)	$105(^{\circ}) \pm 2^{\circ}$	1		
12(b)	263(^{\circ}) correct answer only	1		

Question	Answer	Marks	Part Marks	Guidance
13(a)	Full working must be shown. $(22 \times 55 + 55 \times 110 + 22 \times 110) \times 2 \times \frac{65}{2000}$ $\div 500$ or equivalent $= 1.2\dots$ or 1.3 so 2 tins required (as 1 tin is not enough).	3	Award 2 marks for $(22 \times 55 + 55 \times 110 + 22 \times 110) \times 2 \times \frac{65}{2000}$ or equivalent. or Award 1 mark for $(22 \times 55 + 55 \times 110 + 22 \times 110)$ or better or equivalent or for <i>their</i> surface area $\times \frac{65}{2000}$ or equivalent.	Working must be shown for 2 or 3 marks. Do not accept 629.2 m^l to imply working. Or better for 1 mark, e.g. $(22 \times 55 + 55 \times 110 + 22 \times 110) \times 2$ or 19 360 Accept 19 360 (cm²) without working shown for 1 mark. Alternate for 3 marks, e.g. $(22 \times 55 + 55 \times 110 + 22 \times 110) \times 2 \times \frac{65}{2000} = 629.2$ and 1 tin is 500 m^l so not enough, 2 tins = 1000 m^l so is enough.
13(b)	5.5 (cm by) 2.75 (cm)	1		Both answers correct in either order for the mark.
14	13	1		

Question	Answer	Marks	Part Marks	Guidance
15		2	Award 1 mark for three numbers correctly placed.	
16	A correct answer with geometrical reasons, e.g. • $(f =) 56^\circ$ • $EDC = 56$ as angles on a straight line add up to 180 • $(EDC$ and f are) corresponding angles.	3	Award 1 mark for each bullet point.	Other methods are possible, e.g. • $(f =) 56^\circ$ • $ACD = 124$ as corresponding angle to EDF • $(ACD$ and f) angles on a straight line add up to 180
17	$(m =) \frac{t}{5} + 3$ or $(m =) \frac{t + 15}{5}$	2	Award 1 mark for $\frac{t}{5} = m - 3$ or $t + 15 = 5m$	
18	mode and 5 and median and 4	2	Award 1 mark for mode and 5 or median and 4	Accept reverse only for 1 mark, i.e. median and 4 and mode and 5
19(a)	Parallelogram drawn at (1, 4), (7, 4), (9, 6), (3, 6)	2	Award 1 mark for an enlargement of scale factor 2 with the wrong centre.	
19(b)	(20, 34) (17, 34) (23, 37) (17, 37) (23, 40) (20, 40)	1		Accept any clear indication. Both answers correct for the mark.

Question	Answer	Marks	Part Marks	Guidance
20(a)	13 (seconds) 48 (seconds) 25 (%)	2	Award 1 mark for two correct answers.	
20(b)	A value greater than 35 (seconds)	1		
20(c)	A correct comparison with supporting value, e.g. The median for Class P is 28 (seconds) so Class P are quicker (on average than Class Q to solve the puzzle).	2	Award 1 mark for a correct comparison without supporting value or for finding the median for Class P is 28 (seconds).	Note median for Class P is lower is insufficient for comparison.
21	$(x =) 7.5$	3	Award 1 mark for $20x - 30 = 133 - 2x + 2$ with at most one term incorrect or better. and Award 1 mark for correctly collecting <i>their</i> letters on one side and <i>their</i> numbers on the other, i.e. $20x + 2x = 133 + 2 + 30$ or better.	Or better, e.g. $22x = 165$
22	29.3(m)	4	Award 3 marks for - values which round to 29 300 - values which round to 1.95 or $2 \times \pi \times 311 \times 15 \div 1000$ or 9.33π or Award 2 marks for $2 \times \pi \times 311 \times 15$ or 9330π or for $2 \times \pi \times 311 \div 1000$ or Award 1 mark for $2 \times \pi \times 311$ or 1954 or 622π or for <i>their</i> length in mm $\div 1000$	Implied by a value that rounds to 29.3 Accept conversion to cm for 1 or 2 marks.